ANALYSIS REPORT

Wave 1 High-Order Harmonic Tracking Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Wave 1` high-order harmonic tracking campaign. The campaign tested whether explicit higher-order harmonic bases reduce the over-smoothing observed in the earlier Wave 1 models.

The campaign completed all 18 planned runs with zero launcher failures. The campaign winner by the configured selection policy, minimum `test_mae` with `test_rmse`, `val_mae`, and parameter count as tie breakers, is `te_harmonic_dense360_tracking_Fw`.

Campaign Artifacts

Artifact	Path
Campaign output	`output/training_campaigns/2026-05-20-10-11-06_wave1_high_order_harmonic_tracking_campaign_2026_05_19_17_40_01`
Campaign leaderboard	`output/training_campaigns/2026-05-20-10-11-06_wave1_high_order_harmonic_tracking_campaign_2026_05_19_17_40_01/campaign_leaderboard.yaml`
Best-run pointer	`output/training_campaigns/2026-05-20-10-11-06_wave1_high_order_harmonic_tracking_campaign_2026_05_19_17_40_01/campaign_best_run.yaml`
Execution report	`output/training_campaigns/2026-05-20-10-11-06_wave1_high_order_harmonic_tracking_campaign_2026_05_19_17_40_01/campaign_execution_report.md`
Planning report	`doc/reports/campaign_plans/wave_1/2026-05-19-17-40-01_wave1_high_order_harmonic_tracking_campaign_plan_report.md`
Technical document	`doc/technical/2026-05/2026-05-19/2026-05-19-17-32-08_wave1_high_order_harmonic_tracking.md`

Execution Summary

Field	Value
Campaign name	`wave1_high_order_harmonic_tracking_campaign_2026_05_19_17_40_01`
Started at	`2026-05-20T10:11:06+02:00`
Finished at	`2026-05-20T12:25:49+02:00`
Completed runs	18
Failed runs	0
Tested model types	`harmonic_regression`, `residual_harmonic_mlp`
Tested direction scopes	`global`, `Fw`, `Bw`
Tested harmonic banks	`RCIM sparse`, `0..240`, `0..360`

Campaign Winner

Field	Value
Run name	`te_harmonic_dense360_tracking_Fw`
Run instance	`2026-05-20-10-43-22__te_harmonic_dense360_tracking_fw`
Model family	`harmonic_regression_fw`
Model type	`harmonic_regression`
Direction scope	forward only
Harmonic bank	dense `0..360`
Trainable parameters	4,326
Validation MAE	0.002610
Validation RMSE	0.003057
Test MAE	0.002916
Test RMSE	0.003237

The winning checkpoint is stored at `output/training_runs/harmonic_regression_fw/2026-05-20-10-43-22__te_harmonic_dense360_tracking_fw/checkpoints/harmonic_regression-epoch=053-val_mae=0.00261006.ckpt`.

Leaderboard

Rank	Run	Scope	Harmonics	Test MAE	Test RMSE
1	`te_harmonic_dense360_tracking_Fw`	Fw	`0..360`	0.002916	0.003237
2	`te_harmonic_dense240_tracking_Fw`	Fw	`0..240`	0.002935	0.003239
3	`te_harmonic_rcim_sparse_tracking_Fw`	Fw	RCIM sparse	0.002943	0.003254
4	`te_residual_harmonic_rcim_sparse_tracking_Bw`	Bw	RCIM sparse	0.003042	0.003548
5	`te_residual_harmonic_dense360_tracking_Bw`	Bw	`0..360`	0.003068	0.003545
6	`te_residual_harmonic_rcim_sparse_tracking_Fw`	Fw	RCIM sparse	0.003089	0.003498
7	`te_residual_harmonic_dense240_tracking_global`	global	`0..240`	0.003162	0.003598
8	`te_residual_harmonic_dense240_tracking_Bw`	Bw	`0..240`	0.003188	0.003717
9	`te_residual_harmonic_dense240_tracking_Fw`	Fw	`0..240`	0.003304	0.003773
10	`te_residual_harmonic_rcim_sparse_tracking_global`	global	RCIM sparse	0.003378	0.003902
11	`te_harmonic_dense240_tracking_Bw`	Bw	`0..240`	0.003400	0.003886
12	`te_harmonic_dense360_tracking_Bw`	Bw	`0..360`	0.003403	0.003866
13	`te_harmonic_rcim_sparse_tracking_Bw`	Bw	RCIM sparse	0.003406	0.003894
14	`te_residual_harmonic_dense360_tracking_global`	global	`0..360`	0.003434	0.003957
15	`te_residual_harmonic_dense360_tracking_Fw`	Fw	`0..360`	0.003568	0.004118
16	`te_harmonic_rcim_sparse_tracking_global`	global	RCIM sparse	0.020767	0.022376
17	`te_harmonic_dense360_tracking_global`	global	`0..360`	0.020780	0.022399
18	`te_harmonic_dense240_tracking_global`	global	`0..240`	0.020787	0.022388

Technical Interpretation

The strongest result is concentrated in the forward-only `harmonic_regression` family. Dense `0..360` wins the campaign with test MAE 0.002916, followed very closely by dense `0..240` at 0.002935 and the RCIM sparse bank at 0.002943. The absolute margin between dense `0..360` and RCIM sparse is small, about 0.000027 test MAE, so the larger basis is useful in this campaign but not a dramatic scalar-metric breakthrough.

The result is still direction dependent. The same direct harmonic-regression form is weak in the global bidirectional setting, where all three global harmonic-regression runs remain around 0.02077 test MAE. This confirms that the direct harmonic basis should not be treated as a single global solution without additional conditioning or direction separation.

The residual-harmonic family behaves differently. In backward-only scope, the RCIM sparse residual model has the best test MAE at 0.003042, while dense `0..360` has a slightly lower test RMSE and better validation metrics but a higher test MAE. In global scope, dense `0..240` is the best high-order residual candidate from this campaign at 0.003162 test MAE, but it does not improve the already registered global residual-harmonic best run.

The campaign supports the original hypothesis only partially. Explicit high-order harmonic bases improve the best new forward-only harmonic result and provide a better mechanism for representing fast oscillatory components than a smooth MLP-only predictor. However, scalar MAE and RMSE are not enough to prove that the predicted transmission-error curves visually recover all harmonic detail. The next validation step must be curve-overlay inspection through the TE Curve Verification Pipeline.

Registry Effects

The campaign runner updated the family registries and campaign-level winner artifacts. The most relevant effects are:

Registry Scope	New Relevant Entry	Test MAE	Interpretation
`harmonic_regression_fw`	`te_harmonic_dense360_tracking_Fw`	0.002916	New forward harmonic family best
`harmonic_regression_bw`	`te_harmonic_dense240_tracking_Bw`	0.003400	New backward harmonic family best
`residual_harmonic_mlp_fw`	`te_residual_harmonic_rcim_sparse_tracking_Fw`	0.003089	New forward residual-harmonic candidate
`residual_harmonic_mlp_bw`	`te_residual_harmonic_rcim_sparse_tracking_Bw`	0.003042	New backward residual-harmonic family best
`harmonic_regression`	`te_harmonic_rcim_sparse_tracking_global`	0.020767	Best among weak global harmonic runs

The current program-level winner remains the previously registered `tree_fw` run with test MAE 0.002743. This closeout therefore does not promote a new program-level best model for deployment.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all 18 runs completed, the leaderboard and best-run artifacts exist, and the family registries were refreshed.

From a modeling standpoint, the high-order harmonic feature is worth keeping. The strongest practical candidate from this campaign is `te_harmonic_dense360_tracking_Fw`, with dense `0..240` and RCIM sparse close enough to remain useful lower-complexity comparison points. No model from this campaign should be exported or promoted as the program-level solution until the TE Curve Verification Pipeline curve-overlay plots confirm that the high-frequency TE components are actually tracked rather than merely improving aggregate error.

Recommended Follow-Up

1. Generate TE Curve Verification Pipeline curve overlays for the top three forward harmonic runs: dense `0..360`, dense `0..240`, and RCIM sparse.
2. Compare local oscillation fidelity, not only MAE, against the existing `tree_fw` winner and the earlier Wave 1 smooth predictors.
3. If dense `0..360` visibly tracks the harmonics better, prepare a bounded promotion campaign focused on forward-only harmonic and hybrid candidates.
4. Keep the global harmonic-regression configuration out of promotion paths unless a better conditioned global formulation is introduced.